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Instituto Superior Tecnico --- Universidade de Lisboa

Computer Science and Engineering Department
Lisboa, Portugal

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Dear Members of the Search Committee,

I am writing to express my interest in an Assistant Professor position in the Computer Science and Engineering Department of Instituto Superior Tecnico, particularly in the area of Graphics and Interaction. My research lies in Human--Computer Interaction and interactive systems, especially creator tools, immersive technologies, and systems that support creation and collaboration with advanced computational media.

I hold dual Bachelor's degrees in Systems and Computing Engineering and Electronic Engineering, and an M.Sc. in Computer Science from Universidad de los Andes, Colombia. I completed a cotutelle Ph.D. between Carleton University, Canada, and Université de Lille, France, on programming-based 3D CAD and interactive tools for personal fabrication. I am currently a Postdoctoral Fellow at Carleton University, and a Visiting Researcher in Australia, working on VR/XR interaction, input techniques, haptics, avatar representation, and collaborative immersive systems.

My research program is centered on two complementary directions. The first is interactive tools for 3D modeling, programming-based CAD, and digital fabrication. In this line, I have studied usability and learnability challenges in code-based design environments and proposed bidirectional techniques that connect visual manipulation and code. The second is interaction in immersive systems, where I have investigated haptic feedback, selection techniques, and the effects of representation and input modality on performance and user experience in VR/XR.

Looking ahead, I want to develop this agenda further in two directions: more effective tools for 3D design and digital fabrication, and immersive systems for creation and collaboration. Across both, I am increasingly interested in how AI technologies can support modeling, feedback, guidance, and interaction, while preserving usability, transparency, and meaningful user control. I am also interested in improving how AI systems communicate with people, making their assistance both more effective and easier to understand and evaluate.

I have published in leading venues in HCI and interactive systems, including CHI, UIST, SUI, ISMAR, and VRST. My academic trajectory has been strongly international, with research experience in Colombia (IMAGINE Lab, Universidad de los Andes), France (CREATIS and INRIA, INSA-Lyon and Université de Lille), Canada (Creative Interactions Lab and MARVEL, Carleton University), and Australia (EMVIS, Monash University). I believe this profile would allow me to contribute to Tecnico's research environment, student training, and international collaboration.

Teaching and mentoring are central to my academic work. I have more than 1,500 hours of teaching experience across mathematics, computer science, and engineering, and last year I served as instructor for a graduate course on Data Interaction Techniques. I would be glad to contribute to teaching in Human--Computer Interaction, graphics, interactive systems, immersive technologies, and project-based courses that connect technical development with evaluation and design.

Instituto Superior Tecnico would offer an excellent environment to continue building a research program at the intersection of HCI, 3D design, immersive systems, and emerging AI-supported interaction. Thank you for your consideration. I would be very pleased to discuss my application further.

Sincerely,


J. Felipe GONZALEZ